

Multiple Choice

1. **Answer:** D

Options: A) square; B) hexagon; C) cube; D) tetrahedron

Note: In the diamond structure of elemental carbon, each carbon atom forms four covalent bonds arranged tetrahedrally, resulting in its nearest neighbors being located at the corners of a tetrahedron.

2. **Answer:** D

Options: A) Bosons have symmetric wave functions and obey the Pauli exclusion principle.; B) Bosons have antisymmetric wave functions and do not obey the Pauli exclusion principle.; C) Fermions have symmetric wave functions and obey the Pauli exclusion principle.; D) Fermions have antisymmetric wave functions and obey the Pauli exclusion principle.

Note: Fermions are particles with half-integer spin that are characterized by antisymmetric wave functions, which leads to the Pauli exclusion principle stating that no two fermions can occupy the same quantum state simultaneously.

3. **Answer:** C

Options: A) There are no changes in the internal energy of the system.; B) The temperature of the system remains constant during the process.; C) The entropy of the system and its environment remains unchanged.; D) The entropy of the system and its environment must increase.

Note: In a reversible thermodynamic process, the system and its surroundings can be returned to their original states without any net change in entropy, reflecting the principle that entropy is conserved in such idealized processes.

4. **Answer:** A

Options: A) rate of change of the electric flux through S; B) electric flux through S; C) time integral of the magnetic flux through S; D) rate of change of the magnetic flux through S

Note: The electric displacement current is defined by Maxwell's equations as the term that accounts for the changing electric field in a dielectric, which directly relates to the rate of change of electric flux through a surface S.

5. **Answer:** D

Options: A) magnetic flux through S; B) rate of change of the magnetic flux through S; C) time integral of the magnetic flux through S; D) rate of change of the electric flux through S

Note: The electric displacement current is defined by Maxwell's equations as the term that accounts for the changing electric field in a dielectric, which directly relates to the rate of change of electric flux through a surface S.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The statement is true because, in the observer's reference frame, the time it takes for a meter stick moving at $0.8c$ to pass by is calculated using the relativistic formula for time dilation, which shows that the stick's length contracts, allowing it to pass in 2.5 ns as observed.

7. **Answer:** False

Options: A) True; B) False

Note: The angular magnification of a telescope is determined by the ratio of the focal lengths of the objective and eyepiece lenses, not by the separation distance of the lenses.

8. **Answer:** False

Options: A) True; B) False

Note: The statement is false because for the angular momentum quantum number $l = 2$, the magnetic quantum number m_l can take on five allowed values: -2 , -1 , 0 , $+1$, and $+2$.

9. **Answer:** True

Options: A) True; B) False

Note: The Hall coefficient indicates the type of charge carriers (positive or negative) in a doped semiconductor by revealing the direction of the induced voltage in a magnetic field, which is dependent on the sign of the charge carriers.

10. **Answer:** False

Options: A) True; B) False

Note: The attraction between Cooper pairs in a superconductor is primarily due to phonon-mediated interactions rather than the weak nuclear force.

Long Form Response

11. **Answer:** A beam of electrons can diffract when interacting with a crystal surface under specific conditions, primarily related to the wavelength of the electrons and the lattice spacing of the crystal. For diffraction to occur, the de Broglie wavelength of the electrons must be comparable to the interatomic distances within the crystal, which typically range from 1 to 3 Å. To achieve this condition, electrons need to have a kinetic energy of approximately 10 eV, as this energy corresponds to a wavelength on the order of a few angstroms. At this energy, the electrons possess sufficient momentum to interact with the crystal lattice and produce observable scattering patterns, revealing information about the crystal structure. Thus, both the energy of the electrons and the properties of the crystal are critical for electron diffraction phenomena.

Note: correct because it identifies the crucial relationship between the de Broglie wavelength of electrons and the lattice spacing of the crystal, establishing that a kinetic

energy of about 10 eV allows the electrons' wavelength to match the scale necessary for diffraction to occur.

12. **Answer:** To determine the final voltage across a charged capacitor when two initially uncharged capacitors are connected in series, we start by considering the initial conditions. Assume we have a charged capacitor with voltage V_0 and two uncharged capacitors, each with capacitance C . When connected in series, the total capacitance C_{total} is given by $\frac{1}{C_{total}} = \frac{1}{C} + \frac{1}{C}$, resulting in $C_{total} = \frac{C}{2}$. The charge Q on the charged capacitor remains constant and is equal to $Q = C_{total}V_{final}$. By applying conservation of charge and the relations of voltage, we find that the final voltage across the series connection is $V_{final} = \frac{2V_0}{3}$, illustrating how the voltage divides among the capacitors in series.

Note: correct because it accurately applies the formula for total capacitance in a series circuit, calculates the resulting charge distribution, and demonstrates how the initial voltage across the charged capacitor affects the final voltage across the combined capacitors.